

BRAF V600E Mutation in Malignant Melanoma-A Romanian Research Experience.

PMID: 38541077 | Indexed in PubMed

Background and Objectives: The most common mutation in malignant melanoma (MM) is the single-point mutation of v-raf murine sarcoma viral oncogene homolog B1 (BRAF) oncogene. Our study aims to evaluate BRAF V600E mutation, highlighting its frequency differences in primary versus metastatic MM. **Materials and Methods:** The study group comprised 133 patients diagnosed with MM in several county hospitals of the north-eastern region of Romania who have been assigned for investigation into BRAF V600E mutation in the private medical system. The material consisted of archived formalin-fixed paraffin-embedded (FFPE) blocks. BRAF V600E mutation was identified using the fully automated IdyllaTM BRAF mutation test system. **Results:** Out of the total of 133 cases, 78 cases were primary tumors, while 55 cases were metastatic MMs. Genetic analysis revealed the presence of BRAF V600E mutation in 66 cases (49.62%) and the wild-type genotype in 67 cases (50.37%). We found a statistically significant difference of the mutation frequency according to age ($p = 0.0072$). The mutated genotype was found in 45 cases out of 78 primary MMs (57.69%) and in 21 cases out of 55 secondary MMs (38.18%), with a statistically significant difference in favor of primary tumors ($p = 0.0413$). The correlations between the histopathological types, Clark's level, Breslow index, ulceration, and lymphovascular invasion, respectively, and the mutated genotype were not statistically significant. BRAF V600E mutation was identified in 15 out of 40 secondary tumors with lymph node location (37.5%) and in 6 out of 15 secondary tumors with another location (40%) without statistically significant differences between the mutation frequency and the location of the secondary tumors. **Conclusions:** Our results support MM high genetic heterogeneity, pointing out the relationship between BRAF V600E mutation and several clinicopathological characteristics, in primary and metastatic MMs, stressing the importance of BRAF testing implementation in Romania.

Beyond the BRAFV600E hotspot: biology and clinical implications of rare BRAF gene mutations in melanoma patients.

PMID: 28278349 | Indexed in PubMed

BRAF gene mutations can be found in approximately 50% of melanomas, but the most common BRAF mutation leads to substitution at residue 600 of the protein, from valine to glutamic acid. BRAFV600E occurs in up to 95% of all melanoma cases and can be successfully blocked by using a combination of BRAF- and MEK inhibitors. The wider availability of next-generation sequencing is revealing more non-V600 BRAF mutations, and the clinical implications of these mutations are widely unknown. In this review, we will discuss the biology of the MAPK pathway and the different types of BRAF mutations as well as their effect on MEK activation. Current literature will be reviewed including in vitro data, case reports and case series.

Detection of BRAFV600E Mutation in Melanoma Patients by Digital PCR of Circulating DNA.

PMID: 30676087 | Indexed in PubMed

About 50% of melanomas have the BRAFV600E mutation. This mutation is an attractive therapeutic

target. The aims of our study were to detect BRAFV600E mutations within circulating cell-free DNA in plasma ("liquid biopsy") by a droplet digital PCR (ddPCR) method, and to investigate how well the Breslow-Clark score can be predicted by ddPCR. We analyzed 113 patients with malignant melanoma. ddPCR was performed using the QX200 system (BIO-RAD®, Hercules). All samples were tested in duplicate. Besides the results of the liquid biopsy, we have collected data on gender and age of the patients, as well as the mitotic activity of the tumor; the tumor subtype and localization, and the Breslow-Clark score. The limit of detection (LoD) was determined by the method of Tzonev. The LoD was found to be five events per well. The BRAFV600E mutation was detected in 37 of 113 samples. A moderate predictive accuracy of the Breslow-Clark score can be attained with the mitotic activity and the type of melanoma as the most important predictors. Our results show that ddPCR is a highly sensitive method and could be used for a routine laboratory detection of the BRAFV600E mutation as well as for follow-up monitoring to determine the treatment response in patients with malignant melanomas.

Vemurafenib treatment for patients with locally advanced, unresectable stage IIIC or metastatic melanoma and activating exon 15 BRAF mutations other than V600E.

PMID: 29076950 | Indexed in PubMed

BRAF mutations are found in ~50% of metastatic melanomas, most commonly in codon V600. Vemurafenib improves progression-free survival and overall survival in patients with advanced BRAF-mutated melanoma. The results of a descriptive study evaluating vemurafenib in patients with advanced melanoma harbouring BRAF mutations other than V600E are reported. Eligible patients with stage IIIC or IV melanoma and non-V600E BRAF mutations received vemurafenib (960 mg, twice daily). End points included investigator-assessed best overall response rate (primary), time to response, duration of response, progression-free survival, overall survival and safety. Planned (V600K vs. non-V600K mutations) subgroup analyses were carried out. Thirty-one patients were enrolled; 13 (42%) had V600K mutations and 18 (58%) had other mutations. Investigator-assessed confirmed that the best overall response rate was 23% (95% confidence interval=10-41%) in the overall population, and was similar between patients with V600K mutations (23%; 95% confidence interval=5-54%) versus other mutations (22%; 95% confidence interval=6-48%). Responses were observed in patients with V600K (n=3), V600E2 (n=1), V600R (n=1), L597S (n=1) and D594G (n=1) mutations. No new safety signals were reported. Vemurafenib showed activity in patients with advanced melanoma with rarer BRAF mutations.

Impact of Genomic Mutation on Melanoma Immune Microenvironment and IFN-1 Pathway-Driven Therapeutic Responses.

PMID: 39061208 | Indexed in PubMed

The BRAFV600E mutation, found in approximately 50% of melanoma cases, plays a crucial role in the activation of the MAPK/ERK signaling pathway, which promotes tumor cell proliferation. This study aimed to evaluate its impact on the melanoma immune microenvironment and therapeutic responses, particularly focusing on immunogenic cell death (ICD), a pivotal cytotoxic process triggering anti-tumor immune responses. Through comprehensive in silico analysis of the Cancer Genome Atlas data, we explored the association between the BRAFV600E mutation, immune subtype dynamics, and tumor

mutation burden (TMB). Our findings revealed that the mutation correlated with a lower TMB, indicating a reduced generation of immunogenic neoantigens. Investigation into immune subtypes reveals an exacerbation of immunosuppression mechanisms in BRAFV600E-mutated tumors. To assess the response to ICD inducers, including doxorubicin and Me-ALA-based photodynamic therapy (PDT), compared to the non-ICD inducer cisplatin, we used distinct melanoma cell lines with wild-type BRAF (SK-MEL-2) and BRAFV600E mutation (SK-MEL-28, A375). We demonstrated a differential response to PDT between the WT and BRAFV600E cell lines. Further transcriptomic analysis revealed upregulation of IFNAR1, IFNAR2, and CXCL10 genes associated with the BRAFV600E mutation, suggesting their involvement in ICD. Using a gene reporter assay, we showed that PDT robustly activated the IFN-1 pathway through cGAS-STING signaling. Collectively, our results underscore the complex interplay between the BRAFV600E mutation and immune responses, suggesting a putative correlation between tumors carrying the mutation and their responsiveness to therapies inducing the IFN-1 pathway, such as the ICD inducer PDT, possibly mediated by the elevated expression of IFNAR1/2 receptors.

The effect of a widespread cancer-causing mutation on the inactive to active dynamics of the B-Raf kinase.

PMID: 25868080 | Indexed in PubMed

Protein kinases play a key role in regulating cellular processes. Kinase dysfunction can lead to disease, making them an attractive target for drug design. The B-Raf kinase is a key target for the treatment of melanoma since a single mutation (V600E) is found in more than 50% of all malignant melanomas. Despite the importance of B-Raf in melanoma treatment, the molecular mechanism by which the mutation increases kinase activity remains elusive. Since kinases are tightly regulated by a conformational transition between an active and inactive state, which is difficult to capture experimentally, large-scale enhanced-sampling simulations are performed to examine the mechanism by which the V600E mutation enhances the activity of the B-Raf monomer. The results reveal that the mutation has a twofold effect. First, the mutation increases the barrier of the active to inactive transition trapping B-Raf in the active state. The mutation also increases the flexibility of the activation loop which might speed-up the rate-limiting step of phosphorylation. Both effects can be explained by the formation of salt-bridges with the Glu600 residue.

BRAF mutation testing algorithm for vemurafenib treatment in melanoma: recommendations from an expert panel.

PMID: 23360189 | Indexed in PubMed

The incidence of melanoma has increased rapidly over the past 30 years, and the disease is now the sixth most common cancer among men and women in the U.K. Many patients are diagnosed with or develop metastatic disease, and survival is substantially reduced in these patients. Mutations in the BRAF gene have been identified as key drivers of melanoma cells and are found in around 50% of cutaneous melanomas. Vemurafenib (Zelboraf(®) ; Roche Molecular Systems Inc., Pleasanton, CA, U.S.A.) is the first licensed inhibitor of mutated BRAF, and offers a new first-line option for patients with unresectable or metastatic melanoma who harbour BRAF mutations. Vemurafenib was developed in conjunction with a companion diagnostic, the cobas(®) 4800 BRAF V600 Mutation Test. The purpose of this paper is to

make evidence-based recommendations to facilitate the implementation of BRAF mutation testing and targeted therapy in patients with metastatic melanoma in the U.K. The recommendations are the result of a meeting of an expert panel and have been reviewed by melanoma specialists and representatives of the National Cancer Research Network Clinical Study Group on behalf of the wider melanoma community. This article is intended to be a starting point for practical advice and recommendations, which will no doubt be updated as we gain further experience in personalizing therapy for patients with melanoma.

Reactivation of mitogen-activated protein kinase (MAPK) pathway by FGF receptor 3 (FGFR3)/Ras mediates resistance to vemurafenib in human B-RAF V600E mutant melanoma.

PMID: 22730329 | Indexed in PubMed

Oncogenic B-RAF V600E mutation is found in 50% of melanomas and drives MEK/ERK pathway and cancer progression. Recently, a selective B-RAF inhibitor, vemurafenib (PLX4032), received clinical approval for treatment of melanoma with B-RAF V600E mutation. However, patients on vemurafenib eventually develop resistance to the drug and demonstrate tumor progression within an average of 7 months. Recent reports indicated that multiple complex and context-dependent mechanisms may confer resistance to B-RAF inhibition. In the study described herein, we generated B-RAF V600E melanoma cell lines of acquired-resistance to vemurafenib, and investigated the underlying mechanism(s) of resistance. Biochemical analysis revealed that MEK/ERK reactivation through Ras is the key resistance mechanism in these cells. Further analysis of total gene expression by microarray confirmed a significant increase of Ras and RTK gene signatures in the vemurafenib-resistant cells. Mechanistically, we found that the enhanced activation of fibroblast growth factor receptor 3 (FGFR3) is linked to Ras and MAPK activation, therefore conferring vemurafenib resistance. Pharmacological or genetic inhibition of the FGFR3/Ras axis restored the sensitivity of vemurafenib-resistant cells to vemurafenib. Additionally, activation of FGFR3 sufficiently reactivated Ras/MAPK signaling and conferred resistance to vemurafenib in the parental B-RAF V600E melanoma cells. Finally, we demonstrated that vemurafenib-resistant cells maintain their addiction to the MAPK pathway, and inhibition of MEK or pan-RAF activities is an effective therapeutic strategy to overcome acquired-resistance to vemurafenib. Together, we describe a novel FGFR3/Ras mediated mechanism for acquired-resistance to B-RAF inhibition. Our results have implications for the development of new therapeutic strategies to improve the outcome of patients with B-RAF V600E melanoma.

Evaluation of Heterogeneity in the Coding Region of BRAF, MAP2K1, and MAP2K2 Genes in Primary and Metastatic Melanomas.

PMID: 39588764 | Indexed in PubMed

The incidence of melanoma has been increasing in recent decades. BRAF mutations appear in 50%-70% of melanomas. The BRAF-targeted therapy increased the disease-free survival of patients with metastatic melanoma, but this response may be short, due to several resistance mechanisms, such as the presence of other subclones with mutations. Evaluation of mutations and heterogeneity in the coding region of the BRAF, MAP2K1, and MAP2K2 genes in primary and metastatic melanomas. Twenty-seven samples of primary and metastatic superficial spreading melanoma (SSM) and acral lentiginous melanoma (ALM)

were analyzed for BRAF, MAP2K1, and MAP2K2 mutations using the next-generation sequencing technique. In ALM, the mutation rate found was 50% in the BRAF and MAP2K1 genes and 28.6% in MAP2K2. In the SSM, BRAF was mutated in 76.9%, MAP2K1 in 30.8%, and MAP2K2 in 23.2% of the cases. All samples were formed by distinct tumor subclones in the same lesion. Intertumoral heterogeneity was present between primary and metastatic lesions of ALM in BRAF, MAP2K1, and MAP2K2; the cases of SSM were heterogeneous for BRAF and MAP2K1. We sought to evaluate the mutations in the BRAF, MAP2K1, and MAP2K2 genes, revealing a heterogeneous mutation profile in samples of ALM and SSM.

Current Insights into the Role of BRAF Inhibitors in Treatment of Melanoma.

PMID: 35761499 | Indexed in PubMed

Melanomas represent only 4% of all skin cancers, but their mortality rate is more than 50 % of any other skin cancer. Alteration in genetic and environmental factors are the risk factors for melanoma development. The RAS/RAF/MEK/ERK or Mitogen-activated protein kinase (MAPK) pathway is activated in melanoma. BRAF activation is necessary to govern differentiation, proliferation, and survival. Mutations in BRAF were found in 80-90% of all melanomas. Over 90% of BRAF mutations occur at codon 600, and over 90% of them are BRAFV600E other common mutations are BRAFV600K, BRAFV600R, BRAF V600'E2', and BRAF V600D. Based on α C-helix and DFG motif (α C-helix-IN/DFG-IN), (α C-helix-IN/DFG-OUT), (α C-helix-OUT/DFG-IN) and (α C-helix-OUT/ DFG-OUT) are four structural types of inhibitors for targeting BRAF. Sorafenib, Vemurafenib, Dabrafenib, and Encorafenib are FDAapproved for the treatment of BRAF. Understanding melanoma pathogenesis, RAS/RAF/MEK/ERK or MAPK pathway, and BRAF conformations, mutations, the problems with FDA approved BRAF inhibitors will be important for new drug discovery, modification of existing BRAF barriers to improve target specific action, and prevent increasing response levels while minimizing toxicity.
